

Thomas Lu

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EDUCATION

University of Michigan

Aug. 2026–Present

PhD in Computer Science and Engineering

Advisor: Gabriel Poesia

University of California, Berkeley

May 2026

BA in Computer Science and Physics

EECS Departmental Citation · Highest Distinction · Honors in EECS

EXPERIENCE

Pramaana Labs

May 2026–Aug. 2026

Research Intern, AI and Formal Methods

Song Group, UC Berkeley EECS

Jan. 2025–Jan. 2026

Undergraduate Researcher · Mentor: Jingxuan He

Ranade Group, UC Berkeley EECS

Aug. 2024–May 2026

Undergraduate Researcher · Advisor: Gireeja Ranade

Rivian Automotive

May 2024–Aug. 2024

Software Engineering Intern, Camera Systems

Speech and Computation Lab, UC Berkeley

Aug. 2022–Aug. 2026

Undergraduate Researcher · Advisor: Gasper Begus

Multi-Physics and Circuit Lab, San Jose State University

May 2021–Dec. 2022

Student Researcher · Advisor: Hiu-Yung Wong

PUBLICATIONS & MANUSCRIPTS

* *equal contribution*

- [1] Arnav Mehta*, Chenjun Guo*, **Thomas Lu***, Manooshree Patel, Niels Voss, Rayna Bhattacharyya, Peter Donovan, and Gireeja Ranade. **Formally Verified AI Co-Reasoning: An Education Case Study**
Findings of EMNLP 2026
Workshop: 3rd AI for Math Workshop @ ICML 2026 (poster).
- [2] Chenjun Guo, Manooshree Patel, Arnav Mehta, Rayna Bhattacharyya, **Thomas Lu**, Niels Voss, Bjoern Hartmann, J.D. Zamfirescu-Pereira, Sarah E. Chasins, and Gireeja Ranade. **Bridging the gap to autoformalize free-form human reasoning**
PLATEAU Workshop, 2026
- [3] Timothe Kasriel, **Thomas Lu**, Qinghua Ding, Jingxuan He, and Dawn Song. **UseFor: Usefulness-driven learning of formal mathematics**
MATH-AI: The 5th Workshop on Mathematical Reasoning and AI @ NeurIPS 2025 (poster).
- [4] Manooshree Patel, Rayna Bhattacharyya, **Thomas Lu**, Arnav Mehta, Niels Voss, Narges Norouzi, and Gireeja Ranade. **LeanTutor: A formally-verified AI tutor for mathematical proofs**
EAAI-26: The 16th Symposium on Educational Advances in Artificial Intelligence
Workshop: 2nd AI for Math Workshop @ ICML 2025 (poster).
- [5] Gasper Begus, **Thomas Lu**, and Zili Wang. **Basic syntax from speech: Spontaneous concatenation in unsupervised deep neural networks**
Annual Meeting of the Cognitive Science Society (CogSci 2024), poster presentation.
- [6] Gasper Begus, **Thomas Lu**, Alan Zhou, Peter Wu, and Gopala Anumanchipalli. **CiwaGAN: Informative imitation with articulatory learning**
Annual Meeting of the Linguistic Society of America (LSA 2025), poster presentation.

- [7] **Thomas Lu**, Albert Lu, and Hiu Yung Wong. **Device Image-IV mapping using variational autoencoder for inverse design and forward prediction**
International Conference on Simulation of Semiconductor Processes and Devices (SISPAD), 2023
- [8] Russell A. Poldrack, **Thomas Lu**, and Gasper Begus. **AI-assisted coding: Experiments with GPT-4**
arXiv preprint arXiv:2304.13187, 2023
- [9] Vasu Eranki, **Thomas Lu**, and Hiu Yung Wong. **Comparison of manifold learning algorithms for rapid circuit defect extraction in SPICE-augmented machine learning**
IEEE Workshop on Microelectronics and Electron Devices (WMED), 2022
- [10] **Thomas Lu**, Varada Kanchi, Kashyap Mehta, Sagar Oza, Tin Ho, and Hiu Yung Wong. **Rapid MOSFET contact resistance extraction from circuit using SPICE-augmented machine learning without feature extraction**
IEEE Transactions on Electron Devices, 68(12):6026–6032, 2021

TEACHING EXPERIENCE

CS 170 (Efficient Algorithms and Intractable Problems) <i>Undergraduate Student Instructor</i>	Fall 2025 <i>UC Berkeley EECS</i>
CS 170 (Efficient Algorithms and Intractable Problems) <i>Tutor</i>	Spring 2024–Spring 2025 <i>UC Berkeley EECS</i>

AWARDS AND HONORS

Renaissance Philanthropy AI For Math Fund <i>Grant awarded to the LeanTutor project (PI: Gireeja Ranade)</i>	Sep. 2025
UC Berkeley Electrical Engineering and Computer Sciences <i>Evergreen Undergraduate Research Award</i>	Jun. 2025
UC Berkeley Institute for Cognitive and Brain Sciences <i>Summer Undergraduate Research Award</i>	May 2023
United States Physics Olympiad <i>Silver Medal</i>	May 2022